Prefix-search comparison

DEC-20261030-0c834c · EXP-PSEARCH-3ea9fe · RUN-PSEARCH-c61489

The finite mechanism criterion is met; a total-cost advantage remains unestablished. Of 48 ordinary matched triples,46 produced a checked genuine partial-prefix refutation and fewer complete-model callbacks. All 48 complete projected sets match independently checked reference certificates. The decision recommends separately admitted replication after strategy/accounting refinement; it authorizes no execution or unchanged scaling.

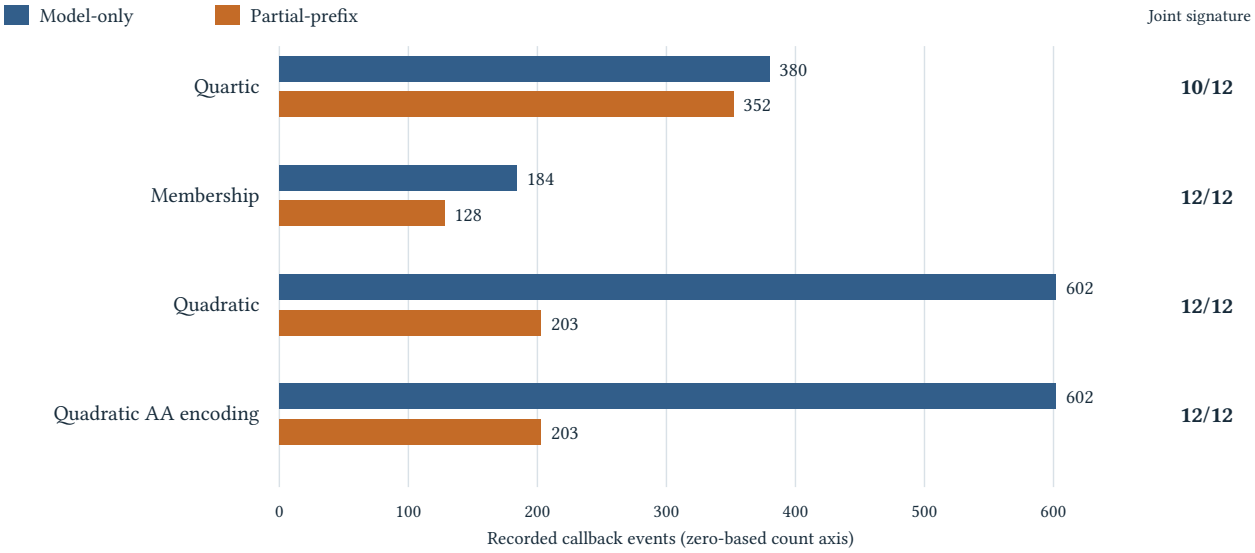
Fixed scope and outcome

The 12 public inputs use F17: $y^2 = x^3 + 2x + 2$ and F97: $y^2 = x^3 + x + 4$, digit widths 2,3 and the first three affine x representatives with least nonzero ordinate. Each retains quartic, membership, quadratic and quadratic_aa encodings, with model-only, partial-prefix and candidate A/A strategies. **144 searches + 9 solver controls + 12 references = 165 executions.** All 153 solver transcripts completed; none was censored or missing.

Cadical195/python-sat1.9.dev17 uses degree 2, 256 admitted rows and 512 PC calls per arm;20000 decisions/conflicts per solve call and frozen outer-call/watchdog limits. Only theory-call timing changes: the candidate also checks eligible contiguous LSB prefixes while assigned_count less than n and depth less than n. The nonlazy propagator and external-clause path remain shared.

Complete-model callback counts

RUN-PSEARCH-c61489 · 12 fixed pairs per encoding · strategy A/A repeat excluded



PC calls:1954⟶1954. Recorded coefficient arithmetic unchanged. No total-work or runtime inference.

The bars sum 12 fixed pairs per encoding; the candidate strategy A/A repeat is excluded. Quadratic_aa is a specified encoding, not an extra replication of this graph. The two ties are c05/quartic (F17,s3,target[5,1]) and c11/quartic (F97,s3,target[4,13]): each 64⟶64 callbacks, with zero partial proofs. These are descriptive event totals, not timing or independent population samples.

5208 identities checked: 5196 ordinary + 12 control. Candidate and strategy A/A have 882 genuine partial proofs each. All 48 ordinary A/A comparisons match exactly; native 51 additionally includes three synthetic classes. Positive-prefix, empty-clause and viable-prefix controls pass; three false multiplier/clause/model mutations are rejected. No search leaf fallback occurred. Twelve references contain 9 projections, repeated across four encodings as 36 per strategy. This certifies projected sets, not all auxiliary roots or scalar recovery.

Fewer model checks do not settle cost

Recorded unit; 48 ordinary rows	Model-only	Partial-prefix
Complete-model callbacks	1768	886
PC calls	1954	1954
Search coefficient multiplications	981664	981664
Search coefficient additions	695814	695814
Evaluation-term visits	87287	46532
Duplicate-prefix visits	3498	8111
Prefix scans	37142	41563

UNKNOWN remains 222, row-capped calls 12, and restriction-term visits 94016 per ordinary strategy. Arithmetic counters omit some primitives; native/Python work, memory, serialization and fsync are not fully counted. Reference instrumentation has separate source-event units. No pooled cost, runtime ranking, new degree-of-regularity result, novelty, formal proof or general ECDLP advantage follows.

Actual independent review and correction

All seven sealed review joints hold. One fresh reviewer separately implemented restriction, multiplier, sparse-coverage, reference-certificate and paired-value checks, sharing frozen polynomial construction. It ran one allowed native verifier invocation and no search replay. Native independence and review-schema gates passed. This is source/transcript/certificate consistency, not universal solver correctness or authenticated callback execution.

Raw-before-export clock/hash ordering is local attestation, not an independent timestamp/archive proof. The pre-export receipt-status correction and other check-development failures are retained. Requested native Codex gpt-6.1-sol/xhigh/fork-none was prospectively admitted; actual serving model/provider/UUID/effective effort remain unverified.

The first export failed because six bundle rows used instance_id=fixed_inventory. A **post-data packaging adapter** changes only that label to protocol instance string0. The reviewer checked all six actual rows: seed 0, replicate 0 and all other fields are unchanged; old labels reproduce all six native errors. The frozen helper and protocol remain unchanged; validation passes and the original failure is retained. The entire export command is not claimed frozen. Six bundles and 786 file entries are packaging units, not observations.

Decision and next action

Recommend a strategy design addressing repeated prefix traversal and avoidable arithmetic calls before a separately frozen held-out comparison. Retain all four encodings, exact identities/projected completeness, A/A/controls, the charged complementary-point baseline and separate setup/search/verification/reconstruction/I/O accounting. **Do not scale the unchanged experiment on callback counts.** No new launch or H/EV/KN/goal transition is authorized; knowledge promotion is not warranted by this single finite observation. Historical PNEXT invalid_output remains unchanged. The general prime-field ECDLP objective remains open.

Sources: sealed review/recompute/attestation in reviews/TASK-20261030-ae3dc7; hashes and actual native receipts in review-inputs.json. Review 12e815fb4ef1abbea4c37dcedf6b48c1814884af; release 6b4d3480db; ledger claim 854ba05c27. Source b2080b14a1; protocol source 3b39d45f4e033f2bc74abc420ef133466fbe65ae; canonical protocol hash b66fb751d2cf5ce1899970a7476df9e5dbaad1661801694204d2dc7d803c271c. PDF inspection and exact ten-path archive remain root-owned pending steps at composition. Current native ledger-record validation governs; the separate legacy JSON decision schema cannot express current hex IDs/knowledge_promotion and is unchanged.